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United States Patent Application Publication

20250281019

Kind Code

A1

Publication Date

September 11, 2025

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NON-RADIATING PUMP COVER

Abstract

A dishwasher pump assembly includes a pump cover configured to reduce the amount of noise transmitted by the cover during operation of the pump. The radiating efficiency of the cover is reduced by reducing the solid percentage of the pump cover. In doing so, the geometry of the pump cover may be modified to inefficient at radiating noise.

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Family ID: 96948134

Appl. No.: 18/596153

Filed: March 05, 2024

Publication Classification

Int. Cl.: A47L15/42 (20060101); F04D29/40 (20060101); F04D29/66 (20060101)

U.S. Cl.:

CPC A47L15/4225 (20130101); F04D29/406 (20130101); F04D29/669 (20130101);

Background/Summary

FIELD OF THE INVENTION

[0001] The present disclosure relates generally to appliances, such as dishwasher appliances, and more particularly to motor covers used in such appliances.

BACKGROUND OF THE INVENTION

[0002] Many appliances utilize fluids for various purposes, such as cleaning purposes, fluid supply purposes, etc. Dishwashers, washing machines, and refrigerators are examples of such appliances. Such appliances typically include conduits for flowing fluids therethrough, both for use in the appliance and for drainage from the appliance. Additionally, pumps may be utilized to encourage fluid flow through such conduits.

[0003] Known pumps utilized with appliances typically include an impeller positioned within a housing through which the fluid flows and an electric motor positioned outside of the housing coupled to the impeller to impart rotation. Pumps can vibrate due to the high rotational speed of the motor and torque pulsations at the impeller. To protect the internal components of the pump from damage or unauthorized access, pumps are often provided with one or more covers. Generally, the covers are substantially flat structures typically formed from a rigid, often fire resistant, plastic material.

[0004] In many cases, the vibration of the pump transmits vibrations to the covers, which act as a drum or speaker diaphragm and amplify the vibration. Consequentially, typical appliance pumps are noisy and, in appliances such as dishwashers, are often the main source of noise for the appliance and consumer dissatisfaction. Accordingly, appliance pumps that transmit less vibration and noise may be desirable. Covers for pumps that are relatively inefficient at transmitting vibration may be particularly useful.

BRIEF DESCRIPTION OF THE INVENTION

[0005] Aspects and advantages of the invention will be set forth in part in the following description, may be apparent from the description, or may be learned through practice of the invention.

[0006] In one exemplary aspect, a dishwashing appliance defining a vertical direction, a lateral direction, and a transverse direction is presented. The dishwashing appliance comprises a tub positioned within a cabinet and defining a wash chamber, a door pivotally mounted to the cabinet to provide selective access to the wash chamber, a rack mounted within the wash chamber and configured for receipt of articles for cleaning, a plurality of spray assemblies and a primary supply conduit is fluidly coupled to the plurality of spray assemblies. A pump assembly is fluidly coupled to the primary supply conduit, the pump assembly comprising a housing supporting an impeller for rotation within the housing, a motor coupled to the impeller to impart rotation, and a cover affixed to the housing, the cover comprising a generally planar panel having an inner surface and an outer surface, the generally planar panel comprising a plurality of apertures formed through the inner surface and the outer surface, wherein the apertures comprise between about 55% and about 75% of a surface area of the generally planar panel.

[0007] In another exemplary aspect an appliance pump assembly comprises a housing supporting an impeller for rotation within the housing, a motor coupled to the impeller to impart rotation, and a cover affixed to the housing, the cover comprising a generally planar panel having an inner surface and an outer surface, the generally planar panel comprising a plurality of apertures formed through the inner surface and the outer surface, wherein the apertures comprise between about 55% and about 75% of a surface area of the generally planar panel.

[0008] These and other features, aspects and advantages of the present invention will become better understood with reference to the following description and appended claims. The accompanying drawings, which are incorporated in and constitute a part of this specification, illustrate embodiments of the invention and, together with the description, serve to explain the principles of the invention.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0009] A full and enabling disclosure of the present invention, including the best mode thereof, directed to one of ordinary skill in the art, is set forth in the specification, which makes reference to the appended figures.

[0010] FIG. 1 provides a perspective view of an exemplary embodiment of a dishwashing appliance of the present disclosure with a door in a partially open position;

[0011] FIG. 2 provides a side, cross-sectional view of the exemplary dishwashing appliance of FIG. 1;

[0012] FIG. 3 provides a perspective view of a pump assembly in accordance with an embodiment of the present disclosure;

[0013] FIG. 4 provides a plan view of a cover in accordance with an embodiment of the present disclosure;

[0014] FIG. 5 provides a plan view of a panel in accordance with an embodiment of the present disclosure;

[0015] FIG. 6 provides a plan view of the cover of FIG. 4 with damping treatment applied in accordance with an embodiment of the present disclosure;

[0016] FIG. 7 provides a sectional view of the cover of FIG. 6 taken along VI-VI; and

[0017] FIG. 8 provides a perspective view of another cover in accordance with an embodiment of the present disclosure.

[0018] Repeated use of reference characters in the present specification and drawings is intended to represent the same or analogous features or elements of the present invention.

DETAILED DESCRIPTION OF THE INVENTION

[0019] Reference now will be made in detail to embodiments of the invention, one or more examples of which are illustrated in the drawings. Each example is provided by way of explanation of the invention, not limitation of the invention. In fact, it will be apparent to those skilled in the art that various modifications and variations can be made in the present invention without departing from the scope or spirit of the invention. For instance, features illustrated or described as part of one embodiment can be used with another embodiment to yield a still further embodiment. Thus, it is intended that the present invention covers such modifications and variations as come within the scope of the appended claims and their equivalents.

[0020] As used herein, the terms “first,” “second,” and “third” may be used interchangeably to distinguish one component from another and are not intended to signify location or importance of the individual components. The terms “includes” and “including” are intended to be inclusive in a manner similar to the term “comprising.” Similarly, the term “or” is generally intended to be inclusive (i.e., “A or B” is intended to mean “A or B or both”). In addition, here and throughout the specification and claims, range limitations may be combined and/or interchanged. Such ranges are identified and include all the sub-ranges contained therein unless context or language indicates otherwise. For example, all ranges disclosed herein are inclusive of the endpoints, and the endpoints are independently combinable with each other. The singular forms “a,” “an,” and “the” include plural references unless the context clearly dictates otherwise.

[0021] Approximating language, as used herein throughout the specification and claims, may be applied to modify any quantitative representation that could permissibly vary without resulting in a change in the basic function to which it is related. Accordingly, a value modified by a term or terms, such as “generally,” “about,” “approximately,” and “substantially,” are not to be limited to the precise value specified. In at least some instances, the approximating language may correspond to the precision of an instrument for measuring the value, or the precision of the methods or machines for constructing or manufacturing the components and/or systems. For example, the approximating language may refer to being within a 10 percent margin, i.e., including values within ten percent greater or less than the stated value. In this regard, for example, when used in the

context of an angle or direction, such terms include within ten degrees greater or less than the stated angle or direction, e.g., “generally vertical” includes forming an angle of up to ten degrees in any direction, e.g., clockwise or counterclockwise, with the vertical direction V.

[0022] The word “exemplary” is used herein to mean “serving as an example, instance, or illustration.” In addition, references to “an embodiment” or “one embodiment” does not necessarily refer to the same embodiment, although it may. Any implementation described herein as “exemplary” or “an embodiment” is not necessarily to be construed as preferred or advantageous over other implementations. Moreover, each example is provided by way of explanation of the invention, not limitation of the invention. In fact, it will be apparent to those skilled in the art that various modifications and variations can be made in the present invention without departing from the scope of the invention. For instance, features illustrated or described as part of one embodiment can be used with another embodiment to yield a still further embodiment. Thus, it is intended that the present invention covers such modifications and variations as come within the scope of the appended claims and their equivalents.

[0023] Turning to the figures, FIGS. **1** and **2** depict an exemplary domestic dishwasher or dishwashing appliance **100** that may be configured in accordance with aspects of the present disclosure. For the particular embodiment of FIGS. **1** and **2**, the dishwasher **100** includes a cabinet **102** (FIG. **2**) having a tub **104** therein that defines a wash chamber **106**. As shown in FIG. **2**, tub **104** extends between a top **107** and a bottom **108** along a vertical direction V, between a pair of side walls **110** along a lateral direction L, and between a front side **111** and a rear side **112** along a transverse direction T. Each of the vertical direction V, lateral direction L, and transverse direction T are mutually perpendicular to one another.

[0024] The tub **104** includes a front opening **114** and a door **116** hinged at its bottom for movement between a normally closed vertical position (shown in FIG. **2**), wherein the wash chamber **106** is sealed shut for washing operation, and a horizontal open position for loading and unloading of articles from the dishwasher **100**. That is, the door **116** is pivotally mounted to the cabinet **102** to provide selective access to the wash chamber **106**. According to exemplary embodiments, dishwasher **100** further includes a door closure mechanism or assembly **118** that is used to lock and unlock door **116** for accessing and sealing wash chamber **106**.

[0025] As best illustrated in FIG. **2**, tub side walls **110** accommodate one or more rack assemblies (three shown) configured for receipt of articles for cleaning. More specifically, guide rails **120** may be mounted to side walls **110** for supporting a lower rack assembly **122**, a middle rack assembly **124**, and an upper rack assembly **126** for sliding motion. As illustrated, upper rack assembly **126** is positioned at a top portion of wash chamber **106** above middle rack assembly **124**, which is positioned above lower rack assembly **122** along the vertical direction V. Each rack assembly **122**, **124**, **126** is adapted for sliding movement on the guide rails **120** between an extended loading position (second position, not shown) in which the rack is positioned at least partially outside the wash chamber **106**, and a retracted position (first position, as shown in FIGS. **1** and **2**) in which the rack is located inside the wash chamber **106**. This is facilitated, for example, by rollers **128** mounted onto rack assemblies **122**, **124**, **126**, respectively. Although guide rails **120** and rollers **128** are illustrated herein as facilitating movement of the respective rack assemblies **122**, **124**, **126**, it should be appreciated that any suitable sliding mechanism or member may be used according to alternative embodiments.

[0026] Some, or all, of the rack assemblies **122**, **124**, **126** are fabricated into lattice structures including a plurality of wires or elongated members **130** (for clarity of illustration, not all elongated members making up rack assemblies **122**, **124**, **126** are shown in FIG. **2**). In this regard, rack assemblies **122**, **124**, **126** are generally configured for supporting articles within wash chamber **106** while allowing a flow of wash fluid to reach and impinge on those articles, e.g., during a cleaning or rinsing cycle.

[0027] Dishwasher **100** further includes a plurality of spray assemblies for urging a flow of water

or wash fluid onto the articles placed within wash chamber **106**. More specifically, as illustrated in FIG. 2, dishwasher **100** includes a lower spray arm assembly **134** disposed in a lower region **136** of wash chamber **106** and above a sump **138** so as to rotate in relatively close proximity to lower rack assembly **122**. Similarly, a mid-level spray arm assembly **140** is located in an upper region of wash chamber **106** and may be located below and in close proximity to middle rack assembly **124**. In this regard, mid-level spray arm assembly **140** may generally be configured for urging a flow of wash fluid up through middle rack assembly **124** and upper rack assembly **126**. Additionally, an upper spray assembly **142** may be located above upper rack assembly **126** along the vertical direction V. In this manner, upper spray assembly **142** may be configured for urging and/or cascading a flow of wash fluid downward over rack assemblies **122**, **124**, and **126**. As further illustrated in FIG. 2, upper rack assembly **126** may further define an integral spray manifold **144**, which is generally configured for urging a flow of wash fluid substantially upward along the vertical direction V through upper rack assembly **126**.

[0028] The various spray assemblies and manifolds described herein may be part of a fluid distribution system or fluid circulation assembly **150** for circulating water and wash fluid in the tub **104**. More specifically, fluid circulation assembly **150** includes an appliance pump assembly, pump assembly **152** for circulating water and wash fluid (e.g., detergent, water, and/or rinse aid) in the tub **104**. Pump assembly **152** may be located within sump **138** or within a machinery compartment located below sump **138** of tub **104**, as generally recognized in the art. Fluid circulation assembly **150** may include one or more fluid conduits or circulation piping for directing water and/or wash fluid from pump assembly **152** to the various spray assemblies and manifolds. For example, as illustrated in FIG. 2, pump assembly **152** may be fluidly coupled to a primary supply conduit **154**, along rear **112** of tub **104**, along the vertical direction V, to supply wash fluid throughout wash chamber **106**.

[0029] As illustrated, primary supply conduit **154** is used to supply pressurized wash fluid to one or more spray assemblies, e.g., to mid-level spray arm assembly **140** and upper spray assembly **142**. However, it should be appreciated that according to alternative embodiments, any other suitable plumbing configuration may be used to supply pressurized wash fluid throughout the various spray manifolds and assemblies described herein. For example, according to another exemplary embodiment, primary supply conduit **154** could be used to provide pressurized wash fluid to mid-level spray arm assembly **140** and a dedicated secondary supply conduit (not shown) could be utilized to provide pressurized wash fluid to upper spray assembly **142**. Other plumbing configurations may be used for providing pressurized wash fluid to the various spray devices and manifolds at any location within dishwasher appliance **100**.

[0030] Each spray arm assembly **134**, **140**, **142**, integral spray manifold **144**, or other spray device may include an arrangement of discharge ports or orifices for directing wash fluid received from pump assembly **152** onto dishes or other articles located in wash chamber **106**. The arrangement of the discharge ports, also referred to as jets, apertures, or orifices, may provide a rotational force by virtue of pressurized wash fluid flowing through the discharge ports. Alternatively, spray arm assemblies **134**, **140**, **142** may be motor-driven, or may operate using any other suitable drive mechanism. Spray manifolds and assemblies may also be stationary. The resultant movement of the spray arm assemblies **134**, **140**, **142** and the spray from fixed manifolds provides coverage of dishes and other dishwasher contents with a washing spray. Other configurations of spray assemblies may be used as well. For example, dishwasher **100** may have additional spray assemblies for cleaning silverware, for scouring casserole dishes, for spraying pots and pans, for cleaning bottles, etc.

[0031] In operation, pump assembly **152** draws wash fluid in from sump **138**, pressurizes the fluid, and pumps it to a diverter assembly **156** which is positioned within sump **138** of dishwasher appliance. Diverter assembly **156** may include a diverter disk (not shown) disposed within a diverter chamber **158** for selectively distributing the pressurized wash fluid to the spray arm

assemblies **134**, **140**, **142** and/or other spray manifolds or devices. For example, the diverter disk may have a plurality of apertures that are configured to align with one or more outlet ports (not shown) at the top of diverter chamber **158**. In this manner, the diverter disk may be selectively rotated to provide wash fluid to the desired spray device.

[0032] According to an exemplary embodiment, diverter assembly **156** is configured for selectively distributing the flow of pressurized wash fluid from pump assembly **152** to various fluid supply conduits, only some of which are illustrated in FIG. 2 for clarity. More specifically, diverter assembly **156** may include three outlet ports (not shown) for supplying pressurized wash fluid to a first conduit for rotating lower spray arm assembly **134**, a second conduit for rotating mid-level spray arm assembly **140**, and a third conduit for spraying upper spray assembly **142**.

[0033] In the illustrative embodiment of FIG. 3, the pump assembly **152** comprises a pump central housing, housing **170**, supporting an impeller (not shown) for rotation within the housing. A motor **172** is adapted to be coupled to the housing **170** such that the motor is coupled to the impeller to impart rotation to the impeller. In particular the rotor, or shaft, of the motor **172** is coupled to the impeller and imparts a torque to urge the impeller to move a volume of fluid, for example wash fluid, through a pump outlet **176**. As illustrated in FIG. 3, a front cover **174** is disposed over a portion of the motor **172** and may be removably fixed to a first side of the housing **170**. The front cover **174** may also facilitate joining of the motor **172** to the housing **170** and/or the impeller.

[0034] A back panel cover, cover **178** is removably affixed to a second side of the housing **170**, opposite the front cover **174**. The cover **178** may be in contact with the housing **170** at a plurality of contact points **196** (3 shown in FIG. 4) which may include through holes **198**. Some of the contact points may be adapted for removably attaching the cover **178** to the housing **170**. In some embodiments, a plurality of fasteners, for example threaded fasteners such as screws, may be used to secure the cover **178** to the housing **170**, the fasteners passing through the through holes **198** to engage with the housing **170**. In some embodiments, at some, or all, contact points, an energy absorbing material **200** (FIG. 4) may separate the cover **178** from the housing **170**, for example to isolate the cover from vibrations generated at the housing **170** or motor **172**.

[0035] Moving to FIGS. 4 and 5, the cover **178** comprises a generally planar panel **180** generally centrally located within the perimeter **182** of the cover **178**. The panel has an inner surface **184** and an opposite outer surface **186** defining a surface area of the panel **180**, the surface area corresponding generally to the planar portion of the cover **178**. In the illustrated embodiment, the generally planar panel **180** defines one or more apertures **188** formed through the panel **180**, that is through the inner surface **184** and the outer surface **186**. In embodiments, the one or more apertures **188** may comprise between about 55% and about 75% of the surface area, or between about 60% and about 70% of the surface area, or about 65% of the surface area.

[0036] In FIG. 4, the apertures **188** are linear slots and may be parallel linear slots as illustrated. In another exemplary embodiment illustrated in FIG. 5, the apertures **188** may comprise a plurality of circular holes **190**. The circular holes **190** may be formed in a plurality of rows and in some embodiments, the rows may be linear parallel rows. As illustrated, the circular holes **190** in adjacent parallel rows may be offset in a direction parallel to the adjacent parallel rows such that the centers of the circular holes in one row do not align with the centers of the circular holes in an adjacent row in a direction perpendicular to the parallel rows. As shown in FIG. 5, the plurality of circular holes **190** may have the same diameter. In other embodiments, the circular holes **190** may have different diameters. Also as shown in FIG. 5, the circular holes are equally spaced in each row and each row is equally spaced from the adjacent row. As such, the circular holes **190** in each of the plurality of rows are equally spaced from an adjacent circular hole. In other embodiments, the circular holes **190** may be spaced differently in each row and adjacent rows may be differently spaced.

[0037] In some embodiments, a damping treatment is applied to a portion of the inner surface **184** of panel **180**. For example, as illustrated in FIGS. 6 and 7, the damping treatment may comprise a

layer of a viscoelastic material **192** applied to the inner surface **184**. The viscoelastic material **192** exhibits both viscous characteristics and elastic characteristics and may be used as a damping material. The damping treatment may also comprise constrained-layer damping. Constrained-layer damping comprises a viscoelastic damping layer **192** applied to a portion of the inside surface **184** and a constraining layer **194** applied to the viscoelastic damping layer **192**. In some embodiments, the constraining layer **194** may be a metallized layer. In other embodiments, the constraining layer may be a fiber layer, for example a woven material, applied to the viscoelastic layer **194**.

[0038] The constraining layer **194** may include apertures **188** corresponding to the apertures formed in the panel **180**. The apertures in the constraining layer may correspond in number, size, and positioning such that the apertures **188** in the panel **180** such that, when the constraining layer is applied, the apertures **188** remain open, or substantially open. In other embodiments, as illustrated in FIG. **6**, the viscoelastic layer **192** is discretely applied to the solid portions of the panel **180**, that is the portions of the panel **180** between the apertures **188**.

[0039] The apertures **188** described above as linear slots and circular holes are provided as examples of apertures, and not to limit the shapes or orientations of apertures. One of ordinary skill in the art will recognize that apertures of different shapes or orientations than those illustrated would have similar beneficial results. For example, FIG. **8** is illustrative of aperture patterns and panel **180** construction in accordance with the present disclosure. FIG. **8** is illustrative of a cover **178** defining a plurality of apertures **188** and circular holes **190**. The apertures **188** and circular holes **190** reduce the solid surface area of the panel **180** to negatively impact the sound transmission efficiency of the cover **178**.

[0040] The dishwasher **100** is further equipped with a controller **160** to regulate operation of the dishwasher **100**. The controller **160** may include one or more memory devices and one or more microprocessors, such as general or special purpose microprocessors operable to execute programming instructions or micro-control code associated with a cleaning cycle. The memory may represent random access memory such as DRAM, or read only memory such as ROM or FLASH. In one embodiment, the processor executes programming instructions stored in memory. The memory may be a separate component from the processor or may be included onboard within the processor. Alternatively, controller **160** may be constructed without using a microprocessor, e.g., using a combination of discrete analog and/or digital logic circuitry (such as switches, amplifiers, integrators, comparators, flip-flops, AND gates, and the like) to perform control functionality instead of relying upon software.

[0041] The controller **160** may be positioned in a variety of locations throughout dishwasher **100**. In the illustrated embodiment, the controller **160** may be located within a control panel area **162** of door **116** as shown in FIGS. **1** and **2**. In such an embodiment, input/output (“I/O”) signals may be routed between the control system and various operational components of dishwasher **100** along wiring harnesses that may be routed through the bottom of door **116**. Typically, the controller **160** includes a user interface panel/controls **164** through which a user may select various operational features and modes and monitor progress of the dishwasher **100**. In one embodiment, the user interface **164** may represent a general purpose I/O (“GPIO”) device or functional block. In one embodiment, the user interface **164** may include input components, such as one or more of a variety of electrical, mechanical or electro-mechanical input devices including rotary dials, push buttons, and touch pads. The user interface **164** may include a display component, such as a digital or analog display device designed to provide operational feedback to a user. The user interface **164** may be in communication with the controller **160** via one or more signal lines or shared communication busses.

[0042] It should be appreciated that the invention is not limited to any particular style, model, or configuration of dishwasher **100**. The exemplary embodiment depicted in FIGS. **1** and **2** is for illustrative purposes only. For example, different locations may be provided for user interface **164**, different configurations may be provided for rack assemblies **122**, **124**, **126**, different spray arm

assemblies 134, 140, 142 and spray manifold configurations may be used, and other differences may be applied while remaining within the scope of the present subject matter.

[0043] This written description uses examples to disclose the invention, including the best mode, and also to enable any person skilled in the art to practice the invention, including making and using any devices or systems and performing any incorporated methods. The patentable scope of the invention is defined by the claims, and may include other examples that occur to those skilled in the art. Such other examples are intended to be within the scope of the claims if they include structural elements that do not differ from the literal language of the claims, or if they include equivalent structural elements with insubstantial differences from the literal language of the claims.

Claims

1. A dishwashing appliance defining a vertical direction, a lateral direction, and a transverse direction, the dishwasher appliance comprising: a tub positioned within a cabinet and defining a wash chamber; a door pivotally mounted to the cabinet to provide selective access to the wash chamber; a rack mounted within the wash chamber and configured for receipt of articles for cleaning; a plurality of spray assemblies; a primary supply conduit fluidly coupled to the plurality of spray assemblies; a pump assembly fluidly coupled to the primary supply conduit, the pump assembly comprising: a housing supporting an impeller for rotation within the housing; a motor coupled to the impeller to impart rotation; and a cover affixed to the housing, the cover comprising a generally planar panel having an inner surface and an outer surface, the generally planar panel defining one or more apertures formed through the inner surface and the outer surface; wherein the one or more apertures comprise between about 55% and about 75% of a surface area of the generally planar panel.
2. The dishwashing appliance of claim 1, wherein the one or more apertures comprise between about 60% and about 70% of the surface area of the generally planar panel.
3. The dishwashing appliance of claim 2, wherein the one or more apertures comprise about 65% of the surface area of the generally planar panel.
4. The dishwashing appliance of claim 1, wherein the one or more apertures comprise circular holes.
5. The dishwashing appliance of claim 4, wherein the circular holes are formed in a plurality of parallel rows.
6. The dishwashing appliance of claim 5, wherein the plurality of parallel rows are linear.
7. The dishwashing appliance of claim 5, wherein the circular holes in adjacent parallel rows are offset in a direction parallel to the adjacent parallel rows such that centers of circular holes in a row do not align with centers in an adjacent row in a direction perpendicular to the rows.
8. The dishwashing appliance of claim 4, wherein the circular holes in each of the plurality of parallel rows are equally spaced from an adjacent circular hole.
9. The dishwashing appliance of claim 1, wherein the apertures comprise a plurality linear slots.
10. The dishwashing appliance of claim 9, wherein the plurality of linear slots are parallel.
11. The dishwashing appliance of claim 1, further comprising a damping treatment applied to a portion of the inner surface.
12. The dishwashing appliance of claim 11, wherein the damping treatment comprises a layer of viscoelastic material.
13. The dishwashing appliance of claim 11, further comprising constrained-layer damping, the constrained-layer damping comprising a viscoelastic damping layer applied to the portion of the inside surface and a constraining layer applied to the viscoelastic damping layer.
14. The dishwashing appliance of claim 13, wherein the constraining layer includes apertures corresponding to the apertures formed in the generally planar panel . . .
15. The dishwashing appliance of claim 13, wherein the constraining layer is a metallized layer.

- 16.** The dishwashing appliance of claim 13, wherein the constraining layer is a fiber layer.
- 17.** An appliance pump assembly comprising: a housing supporting an impeller for rotation within the housing; a motor coupled to the impeller to impart rotation; and a cover affixed to the housing, the cover comprising a generally planar panel having an inner surface and an outer surface, the generally planar panel defining one or more apertures formed through the inner surface and the outer surface; wherein the one or more apertures comprise between about 55% and about 75% of a surface area of the generally planar panel.
- 18.** The appliance pump assembly of claim 17, wherein the one or more apertures comprise about 65% of the surface area of the generally planar panel.
- 19.** The appliance pump of claim 17, wherein the one or more apertures comprise circular holes formed in a plurality of parallel rows.
- 20.** The appliance pump of claim 17, further comprising a damping treatment applied to a portion of the inner surface, wherein the damping treatment comprises a layer of viscoelastic material.
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